Abhinav Pathak

Full-Stack AI Engineer

Greater Noida, Uttar Pradesh · permanent: Arrah, Bihar

Email pathakabhinav0405@gmail.com
Portfolio abhinavpathak.vercel.app
GitHub github.com/pathakabhi-04
LinkedIn linkedin.com/in/abhinav-pathak-882733343

SUMMARY

B.Tech Computer Science student who builds end-to-end AI systems in computer vision and retrieval-augmented generation, from model training and evaluation to a served API. Core coding in Python and C++. Looking for an AI engineering role.

EDUCATION

Qualification	Institution / Board	Year	Score
B.Tech, Computer Science Engineering	Bennett University	2024 – 2028	CGPA 8.43 / 10 (current)
Class XII	CBSE	2023	80%
Class X	CBSE	2021	96.4%

SKILLS

Languages	Python, C++, Bash
Core CS	Data Structures & Algorithms, Operating Systems
AI / ML	PyTorch, Ultralytics YOLO, OpenCV, Hugging Face Transformers, sentence-transformers, LangChain, LangGraph, RAG, RAGAS, Ollama, Groq
Backend & tools	FastAPI, Pydantic, ChromaDB, FAISS, NumPy, pandas, scikit-learn, Git, Linux, pytest, Streamlit, Vercel

PROJECTS

DermaSense · PyTorch, YOLO11, U-Net, FastAPI 2026
Skin-lesion analysis system that checks photo quality, finds and outlines lesions, and tracks how they change between visits. I built the eight-stage pipeline and its FastAPI service (0.52 s per image on CPU), and ran experiments that exposed a data gap, a patient leak and a bug inflating the scores.

TemporalRAG · YOLOv8m, ByteTrack, LLaVA-7B, ChromaDB 2026
Natural-language search over CCTV and dashcam video, with answers that cite exact timestamps. I built the pipeline from detection and tracking (YOLOv8m fine-tuned on Indian roads) to hybrid retrieval, as a two-person team in three weeks.

RAG Document QA · BM25, ChromaDB, cross-encoder, RAGAS 2026
Conversational question answering over multiple PDFs, citing the source page. I designed hybrid retrieval with reranking and a confidence gate that refuses weak answers, and evaluated it on a 278-question hand-verified dataset.

iTantra · Kotlin, ONNX Runtime, IndicTrans2 2026
Offline Android app that relays speech between phones with no internet, for ISRO's Smart India Hackathon problem. I built on-device translation in four languages, cutting its memory from 999 MB to 322 MB, plus voice clips, photo sharing, roles and blood requests.

MeshDrive · Rust, Tauri, libp2p, BLAKE3 2026
Peer-to-peer cloud drive that splits files into hashed chunks stored on users' own machines. Team project; I worked across the backend: chunking, the REST API and peer-to-peer networking.

Stocklite · FastAPI, Llama 3.2 (LoRA), NewsAPI 2026
AI stock-analysis chat assistant built on technical indicators, news and a fine-tuned Llama model. Team project; I built its conversational layer: session memory, follow-up questions and intent routing.

Other projects
ADAS Dashcam Safety (vehicle and pedestrian tracking with a time-to-collision risk score), Semantic Document Search (FAISS), AI Resume Analyzer API (FastAPI, LangGraph).

ACHIEVEMENTS

- **Smart India Hackathon 2026: team ranked 7th in the Bennett University internal round with iTantra (ISRO problem statement SIH26173).**